

Stat 707 HW3

1. $y_i | \beta \sim N(x_i \beta, \sigma_0^2)$, $i=1, \dots, m$. σ_0^2 known, $\theta = \beta$ scalar, $p=1$

a) obtain a closed form expression for CPO

$$\rightarrow f(y_i | y_{-i}) = \int \underbrace{f(x_i | \beta)}_{\text{likelihood}} \underbrace{p(\beta | y_{-i})}_{\text{flat prior}} d\beta$$

$$\frac{1}{\sigma_0 \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma_0^2} (y_i - x_i \beta)^2 \right\} \quad f(y_{-i} | \theta) = \frac{1}{\sigma_0 \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma_0^2} \left(\sum_{j=1}^m x_j^2 - \beta^2 \sum_{j=1}^m x_j^2 - 2\beta \sum_{j=1}^m x_j y_j \right) \right\}$$

$$\rightarrow \frac{-\sum x_j^2}{2\sigma_0^2} \left(\frac{\sum x_j^2}{\sum x_j^2} \beta^2 - 2\beta \frac{\sum x_j y_j}{\sum x_j^2} \right)$$

$$= \frac{1}{\sigma_0 \sqrt{2\pi}} \exp \left\{ -\frac{\sum x_j^2}{2\sigma_0^2} \left(\beta - \hat{\beta}_{-i} \right)^2 \right\}$$

→ integrating out β we have

$$\left(\sum_{j=1}^m x_j^2 \right)^{-1/2} \quad \text{Normalizing constant}$$

$$\text{so } p(\beta | y_{-i}) = \frac{\sqrt{\sum_{j=1}^m x_j^2}}{\sigma \sqrt{2\pi}} \exp \left\{ -\frac{\sum x_j^2}{2\sigma_0^2} (\beta - \hat{\beta}_{-i})^2 \right\}$$

times

returning, the CPO

$$f(y_i | y_{-i}) = \frac{\sqrt{\sum_{j=1}^m x_j^2}}{2\pi \sigma_0^2} \exp \left\{ -\frac{1}{2\sigma_0^2} \left[(y_i - x_i \beta)^2 + \sum x_j^2 (\beta - \hat{\beta}_{-i})^2 \right] \right\} d\beta$$

$$\boxed{\begin{aligned} ax^2 + bx + c &= 0 \\ \rightarrow a(x + \frac{b}{2a})^2 + e &= 0 \\ c &= \frac{b^2}{4a} \quad e = c - \frac{b^2}{4a} \end{aligned}}$$

$$\rightarrow y_i^2 - 2y_i x_i \beta + x_i^2 \beta^2 + \sum x_j^2 \beta^2 - 2 \sum x_j^2 \hat{\beta}_{-i} \beta + \sum x_j^2 \hat{\beta}_{-i}^2$$

Complete the square...

$$\rightarrow \beta^2 (x_i^2 + \sum x_j^2) - 2\beta (y_i x_i + \sum x_j^2 \hat{\beta}_{-i}) + y_i^2 + \sum x_j^2 \hat{\beta}_{-i}^2$$

$$\rightarrow (x_i^2 + \sum x_j^2) \left(\beta - \frac{y_i x_i + \sum x_j^2 \hat{\beta}_{-i}}{x_i^2 + \sum x_j^2} \right)^2 - \frac{(y_i x_i + \sum x_j^2 \hat{\beta}_{-i})^2}{x_i^2 + \sum x_j^2} + y_i^2 + \sum x_j^2 \hat{\beta}_{-i}^2$$

$$\star \int_{-\infty}^{\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2} (x + \mu)^2} dx = \frac{1}{\sigma \sqrt{2\pi}} \leftrightarrow \beta \sim N \left(\frac{y_i x_i + \sum x_j^2 \hat{\beta}_{-i}}{x_i^2 + \sum x_j^2}, \frac{\sigma_0^2}{x_i^2 + \sum x_j^2} \right)$$

$$f(y_i | y_{-i}) = \frac{\sqrt{\sum_{j=1}^m x_j^2}}{2\pi \sigma_0^2} \frac{\sigma_0^2 \sqrt{2\pi}}{x_i^2 + \sum x_j^2} \exp \left\{ -\frac{1}{2\sigma_0^2} \left(\frac{(y_i x_i + \sum x_j^2 \hat{\beta}_{-i})^2}{x_i^2 + \sum x_j^2} + y_i^2 + \sum x_j^2 \hat{\beta}_{-i}^2 \right) \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{\sqrt{\sum_{j=1}^m x_j^2}}{x_i^2 + \sum x_j^2} \exp \left\{ -\frac{1}{2\sigma_0^2} \left(\frac{(y_i x_i + \sum x_j^2 \hat{\beta}_{-i})^2}{x_i^2 + \sum x_j^2} + y_i^2 + \sum x_j^2 \hat{\beta}_{-i}^2 \right) \right\}$$

Ctd.

$$\frac{-(y_i y_i - \sum x_j^2 \beta_{-1}^2)^2}{x_i^2 - \sum x_j^2} + \frac{(y_i^2 - \sum x_j^2 \beta_{-1}^2)(x_i^2 - \sum x_j^2)}{x_i^2 - \sum x_j^2}$$

$y_i x_i$	$\sum x_j^2 \beta_{-1}^2$
$y_i^2 x_i$	$y_i x_i \sum x_j^2 \beta_{-1}^2$
$\sum x_j^2 \beta_{-1}^2$	$\beta_{-1}^2 (\sum x_j^2)^2$

x_i^2	$\sum x_j^2 \beta_{-1}^2$
$y_i^2 x_i$	$x_i^2 \sum x_j^2 \beta_{-1}^2$
$\sum x_j^2$	$\beta_{-1}^2 (\sum x_j^2)^2$

$$\rightarrow \frac{1}{x_i^2 - \sum x_j^2} \left(-y_i^2 x_i^2 - 2y_i x_i \sum x_j^2 \beta_{-1}^2 + \beta_{-1}^2 (\sum x_j^2)^2 + x_i^2 x_i^2 + x_i^2 \sum x_j^2 \beta_{-1}^2 + y_i^2 \sum x_j^2 - \beta_{-1}^2 (\sum x_j^2)^2 \right)$$

$$\rightarrow \frac{1}{x_i^2 - \sum x_j^2} \left(y_i^2 (\sum x_j^2) - 2y_i x_i (\sum x_j^2 \beta_{-1}^2) + x_i^2 \sum x_j^2 \beta_{-1}^2 \right)$$

$$\rightarrow \frac{1}{x_i^2 - \sum x_j^2} \left(\sum x_j^2 (y_i - x_i \beta_{-1}^2)^2 + \frac{x_i^2 \beta_{-1}^2 (\sum x_j^2)^2}{\sum x_j^2} + x_i^2 \sum x_j^2 \beta_{-1}^2 \right)$$

$$\rightarrow \frac{1}{x_i^2 - \sum x_j^2} \left(\sum x_j^2 (y_i - x_i \beta_{-1}^2)^2 + 2x_i^2 \beta_{-1}^2 \sum x_j^2 \right) = \frac{\sum x_j^2}{x_i^2 - \sum x_j^2} \left((y_i - x_i \beta_{-1}^2)^2 + 2x_i^2 \beta_{-1}^2 \right)$$

$$ax^2 + bx + c = 0$$

$$\rightarrow a(x + \frac{b}{a})^2 + c - \frac{b^2}{4a} = 0$$

$$c = \frac{b^2}{4a}, e = -\frac{b}{2a}$$

$$P(y_i | y_{-i}) = \frac{1}{\sqrt{2\pi}} \frac{\sum x_j^2}{x_i^2 - \sum x_j^2} \exp \left\{ -\frac{1}{2\sigma^2} \left[\frac{\sum x_j^2}{x_i^2 - \sum x_j^2} \left((y_i - x_i \beta_{-1}^2)^2 + 2x_i \beta_{-1}^2 \right) \right] \right\}$$

$$\rightarrow y_i | y_{-i} \sim N \left(\frac{x_i \sum x_j y_j}{\sum x_j^2}, \sigma^2 \left(\frac{x_i^2 - \sum x_j^2}{\sum x_j^2} \right) \right), \hat{\beta}_{-1} = \frac{\sum x_j y_j}{\sum x_j^2}$$

1b) obtain closed form for LDD residuals and standardized residuals

$$r_i = y_i - E(y_i | y_{-i}) = y_i - \frac{x_i \sum x_j y_j}{\sum x_j^2}$$

$$= y_i - x_i \hat{\beta}_{-1}$$

$$\text{standardized: } \frac{r_i}{\sqrt{\text{Var}(y_i | y_{-i})}}$$

$$\frac{y_i - \frac{x_i \sum x_j y_j}{\sum x_j^2}}{\sqrt{\frac{(x_i^2 - \sum x_j^2)}{\sum x_j^2}}}$$

1c) Is it possible to write the previous results in terms of $\hat{\beta}_{OLS}$ or β ?

$$\hat{\beta}_{OLS} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\sum_{j \neq i} x_j y_j + x_i y_i}{\sum x_j^2 + x_i^2} = \frac{\sum x_j y_j}{\sum x_j^2} + \frac{x_i y_i}{\sum x_j^2 + x_i^2} = \hat{\beta}_{-1} + \frac{x_i y_i}{\sum x_j^2 + x_i^2}$$

$$\text{So replace the } \hat{\beta}_{-1} \text{ in a) with } \hat{\beta}_{OLS} \left(1 + \frac{x_i^2}{\sum x_j^2} \right) - \frac{x_i y_i}{\sum x_j^2}$$

2. Find an analytic expression for $\log L_{OLS}$

$$\rightarrow \sum_{i=1}^n \log P(y_i | y_{-i}) = \sum_{i=1}^n \log \left[\frac{1}{\sqrt{2\pi}} \frac{\sum x_j^2}{x_i^2 - \sum x_j^2} \right] + \sum_{i=1}^n \left[-\frac{1}{2\sigma^2} \left[\frac{\sum x_j^2}{x_i^2 - \sum x_j^2} \left((y_i - x_i \hat{\beta}_{-1}^2)^2 + 2x_i \beta_{-1}^2 \right) \right] \right]$$